

Section 5.3, 8.1 - Orthogonality & Gram-Schmidt Algorithm

Recall :

dot product, $v \cdot w = v_1 w_1 + v_2 w_2 + \dots + v_n w_n$

length of vector, $\|v\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$
 $= \sqrt{\|v\|^2}$
 $= \sqrt{v \cdot v}$

θ between v & w , $v \cdot w = \|v\| \|w\| \cos \theta$
 $\cos \theta = \frac{v \cdot w}{\|v\| \|w\|}$

if $v \cdot w = 0 \Rightarrow$ perpendicular / orthogonal

Orthogonal & Orthonormal Sets :

for non-zero vectors in set $S = \{x_1, x_2, \dots, x_k\}$ in $\mathbb{R}^n$...

Orthogonal set $\Rightarrow$ if any pair of vectors is orthogonal

$$x_i \cdot x_j = 0, \quad i \neq j$$

Orthonormal set $\Rightarrow$ if orthogonal & each vector length is 1

$$\|x_i\| = 1, \quad \text{for } x_i \in S$$

→ non-zero Orthogonal sets $\Rightarrow$ linearly independent

→ if we want to make orthogonal into orthonormal...

Normalize each vector, $u = \frac{v}{\|v\|}$

e.g.

verify set $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} \right\}$ is orthogonal, is it orthonormal?

$$\text{check: } \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} = 0$$

$$(1)(-1) + (2)(1) + (1)(-1) = 0$$

$$-1 + 2 - 1 = 0$$

$$0 = 0$$

$$0 = 0 \quad \checkmark$$

check orthonormal:

$$\left\| \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \right\| = \sqrt{1^2 + 2^2 + 1^2} \neq 1 \quad \times$$
$$= \sqrt{6}$$

$$\left\| \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} \right\| = \sqrt{(-1)^2 + 1^2 + (-1)^2} \neq 1 \quad \times$$
$$= \sqrt{1 + 1 + 1}$$
$$= \sqrt{3}$$

$\Rightarrow$ it is orthogonal, but not orthonormal

Orthogonal Bases:

Orthogonal Basis $\Rightarrow$ if it's an orthogonal set

Orthonormal Basis $\Rightarrow$ if it's an orthonormal set

$\Rightarrow$ let $\{f_1, f_2, \dots, f_k\}$ be an orthogonal basis for U subspace...

$$x = \sum \frac{x \cdot f_i}{\|f_i\|^2} (f_i)$$

$\leftarrow \sum \text{proj}_{f_i}(x)$

called Fourier Coefficients.

$\rightarrow$ finds linear combination using orthog. basis

e.g.

The set $\{f_1, f_2\} = \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right\}$ is an orthogonal

basis for $U = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}$

express vector $x = \begin{bmatrix} 12 \\ -6 \\ -6 \end{bmatrix}$ as a linear combination

of f_1 & f_2

$$x = \begin{bmatrix} 12 \\ -6 \\ -6 \end{bmatrix} = \sum \frac{x \cdot f_i}{\|f_i\|^2} (f_i)$$

$$\begin{bmatrix} 12 \\ -6 \\ -6 \end{bmatrix} = \frac{x \cdot f_1}{\|f_1\|^2} (f_1) + \frac{x \cdot f_2}{\|f_2\|^2} (f_2)$$

$$= \frac{\begin{bmatrix} 12 \\ -6 \\ -6 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}}{\left\| \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \right\|^2} (f_1) + \frac{\begin{bmatrix} 12 \\ -6 \\ -6 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}}{\left\| \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right\|^2} (f_2)$$

$$= \frac{18}{2} (f_1) + \frac{18}{6} (f_2)$$

$$\begin{bmatrix} 12 \\ -6 \\ -6 \end{bmatrix} = 9f_1 + 3f_2$$

Gram - Schmidt Algorithm :

Transforms any basis into an Orthogonal basis

let $\{x_1, x_2, \dots, x_k\}$ be a basis for subspace U .
we define vectors $f_1, f_2, \dots, f_k$ as ...

$$f_1 = x_1$$

$$f_2 = x_2 - \text{proj}_{f_1}(x_2)$$

$$f_3 = x_3 - \text{proj}_{f_1}(x_3) - \text{proj}_{f_2}(x_3)$$

$\vdots$

$$f_k = x_k - \text{proj}_{f_1}(x_k) - \text{proj}_{f_2}(x_k) - \dots - \text{proj}_{f_{k-1}}(x_k)$$

$$= x_k - \left[\sum_{i=1}^{k-1} \frac{x \cdot f_i}{\|f_i\|^2} (f_i) \right]$$

$$\Rightarrow \{f_1, f_2, \dots, f_k\} = \text{orthogonal basis for } U$$

e.g.

Apply Gram-Schmidt Alg. to basis $\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}$ for

$$U = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x + y + z = 0 \right\}$$

$$f_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$\begin{aligned} f_2 &= x_2 - \text{proj}_{f_1}(x_2) \\ &= \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} - \left(\frac{\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}}{\left\| \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \right\|^2} \right) (f_1) \\ &= \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} - \begin{bmatrix} 1/2 \\ -1/2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \\ -1 \end{bmatrix} \end{aligned}$$

$$\text{Orthogonal basis} = \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1/2 \\ 1/2 \\ -1 \end{bmatrix} \right\}$$

check orthogonality:

$$\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -1 \end{bmatrix} = (\frac{1}{2}) - (\frac{1}{2}) + 0 = 0 \quad \checkmark$$

e.g.

find an orthogonal basis for $U = \left\{ \begin{bmatrix} x \\ y-x \\ z-y \end{bmatrix} : x, y, z \in \mathbb{R} \right\}$

$\Rightarrow$ first we need to find basis for U , suggest $\text{im}(A)$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\beta = \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$\Rightarrow$ Now we can find orthogonal basis...

$$f_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$f_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - \left[\frac{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}}{\left\| \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \right\|^2} (f_1) \right]$$

$$= \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - \frac{-1}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$f_3 = x_3 - \dots = \frac{1}{3} \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix}$$

$$\Rightarrow \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \frac{1}{3} \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix} \right\}$$

verify by computing $f_1 \cdot f_2$, $f_1 \cdot f_3$, $f_2 \cdot f_3$

Orthogonal Complements:

$$\{0\}^\perp = \mathbb{R}^n$$

$$(\mathbb{R}^n)^\perp = \{0\}$$

Orthogonal Complement of U is ...

$U^\perp = \{v \in \mathbb{R}^n : v \cdot u = 0, \text{ for all } u \in U\}$

U -perp $\rightarrow$

$\rightarrow$ Line & Plane relationship: line's direction vector is $\perp$ to plane's normal

$\rightarrow U^\perp$ can be expressed as null space of a matrix from spanning vectors of U

e.g.

Match each subspace from left-hand column with its orthogonal complement in right-hand column (each is either plane or line)

$$U = \left\{ t \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} : t \in \mathbb{R} \right\}$$

$$X = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x - y + z = 0 \right\}$$

$$V = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : 2x - 3y + z = 0 \right\}$$

$$Y = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x + y - z = 0 \right\}$$

$$W = \left\{ t \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} : t \in \mathbb{R} \right\}$$

$$Z = \left\{ t \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} : t \in \mathbb{R} \right\}$$

$$U \rightarrow U^\perp : \text{plane eq'n } x + y - z = 0 \rightarrow Y$$

$$V \rightarrow V^\perp : \text{line spanned by } [2 \ -3 \ 1]^T \rightarrow Z$$

$$W \rightarrow W^\perp : \text{plane eq'n } x - y + z = 0 \rightarrow X$$

e.g.

let $U = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix} \right\}$. Find a system of homog.

eq'ns that defines $U^\perp$

(i.e. express $U^\perp$ as null space of some matrix)

from
def'n

$$U^\perp = \left\{ \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \in \mathbb{R}^4 : \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = 0 \text{ and } \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \cdot \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix} = 0 \right\}$$

$$= \left\{ \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \in \mathbb{R}^4 : x + 2y + 3z + 4w = 0 \text{ AND } 5x + 6y + 7z + 8w = 0 \right\}$$

so set $U^\perp$ is set of all sol'tns following the homog. system of eq'tns

$$x + 2y + 3z + 4w = 0$$

$$5x + 6y + 7z + 8w = 0$$

and express as $\text{null}(A)$, where $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$

e.g.

Match each subspace on left with its orthogonal complement on the right...

$$\textcircled{1} \text{ span } \left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix} \right\} \longrightarrow \text{null} \begin{bmatrix} 0 & 1 & 1 & -2 \\ 2 & -3 & 0 & 1 \end{bmatrix}$$

$$\textcircled{2} \text{ span } \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ -3 \\ 5 \end{bmatrix} \right\} \longrightarrow \text{null} \begin{bmatrix} 1 & 0 & 0 & -3 \\ 1 & 1 & 2 & 1 \\ 3 & 1 & -3 & 5 \end{bmatrix}$$

$$\textcircled{3} \text{ span } \left\{ \begin{bmatrix} 0 \\ -2 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} -3 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\} \longrightarrow \text{null} \begin{bmatrix} 0 & -2 & 1 & -2 \\ -3 & 2 & 0 & 1 \end{bmatrix}$$

$$\textcircled{4} \text{ span } \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ -3 \end{bmatrix} \right\} \longrightarrow \text{null} \begin{bmatrix} 1 & 1 & 0 & -3 \\ 2 & 1 & 1 & 2 \\ 1 & 3 & 5 & -3 \end{bmatrix}$$

Orthogonal Projection :

let U be subspace w/ basis $\{f_1, f_2, \dots, f_k\}$, Orthog. projection onto U is...

$$\text{proj}_U(x) = \sum \frac{x \cdot f_i}{\|f_i\|^2} (f_i)$$

→ properties, let $p = \text{proj}_U(x)$:

- $p \in U$ and $(x - p) \in U^\perp$
- p is the closest vector to x

e.g.

let $U = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}$. Then $\{f_1, f_2\} = \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right\}$ is an orthogonal basis of U .

Compute orthogonal proj. of $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ onto U .

$$\begin{aligned} \text{proj}_U(x) &= \sum \frac{x \cdot f_i}{\|f_i\|^2} (f_i) \\ &= \frac{x \cdot f_1}{\|f_1\|^2} (f_1) + \frac{x \cdot f_2}{\|f_2\|^2} (f_2) \\ &= \frac{-1}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + \frac{-3}{6} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \\ &= \dots \\ &= \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \end{aligned}$$

Dimension Relationship :

we define $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $T(x) = \text{proj}_U(x)$, then ...

1. T is linear Transformation
2. $\text{im}(A_T) = U$ and $\text{null}(A_T) = U^\perp$
3. $\dim(U) + \dim(U^\perp) = n$